



Inotify is a Linux kernel feature that provides an event-driven framework for applications to monitor changes in the filesystem. This article demonstrates how to use the Inotify API to register for interesting events, and do something in response to those events.

The changes that you can track with *Inotify* could be the creation, deletion, or modification of directories and files. To use the *Inotify* API, you just need to include the header `sys/inotify.h` in your C program.

Watches

A *watch* is at the core of *Inotify*. You use a watch to specify the directory or file you are interested in monitoring for events, and the events that you are interested in. For example, the watch function `inotify_add_watch(fd, root, IN_CREATE | IN_MODIFY | IN_DELETE)` specifies that you want to monitor the directory specified in the string `root`, and you want to monitor all file and directory creation, modification and deletion events, as specified by the watch mask, `IN_CREATE | IN_MODIFY | IN_DELETE` in this directory. The valid watch masks that can be specified are found in `linux/inotify.h`.

The return value of the above function is an integer that uniquely identifies this watch, and on error, a value of -1 is returned. `fd` is the file descriptor that specifies the current

Inotify instance. A watch is uniquely identified by the pathname it is watching, so if you want to specify a different watch for the same path, you will have to call `inotify_add_watch()` with the same path, but with the new combination of events that you want to watch.

A simple notification program

Listing 1 shows a simple demo *Inotify* application, which watches a directory for file and directory creation, modification and deletion events. See *References* for the source that is available for download.

Listing 1:

```
#include <stdio.h>
#include <stdlib.h>
#include <errno.h>
#include <sys/types.h>
#include <sys/inotify.h>
#include <limits.h>
```